

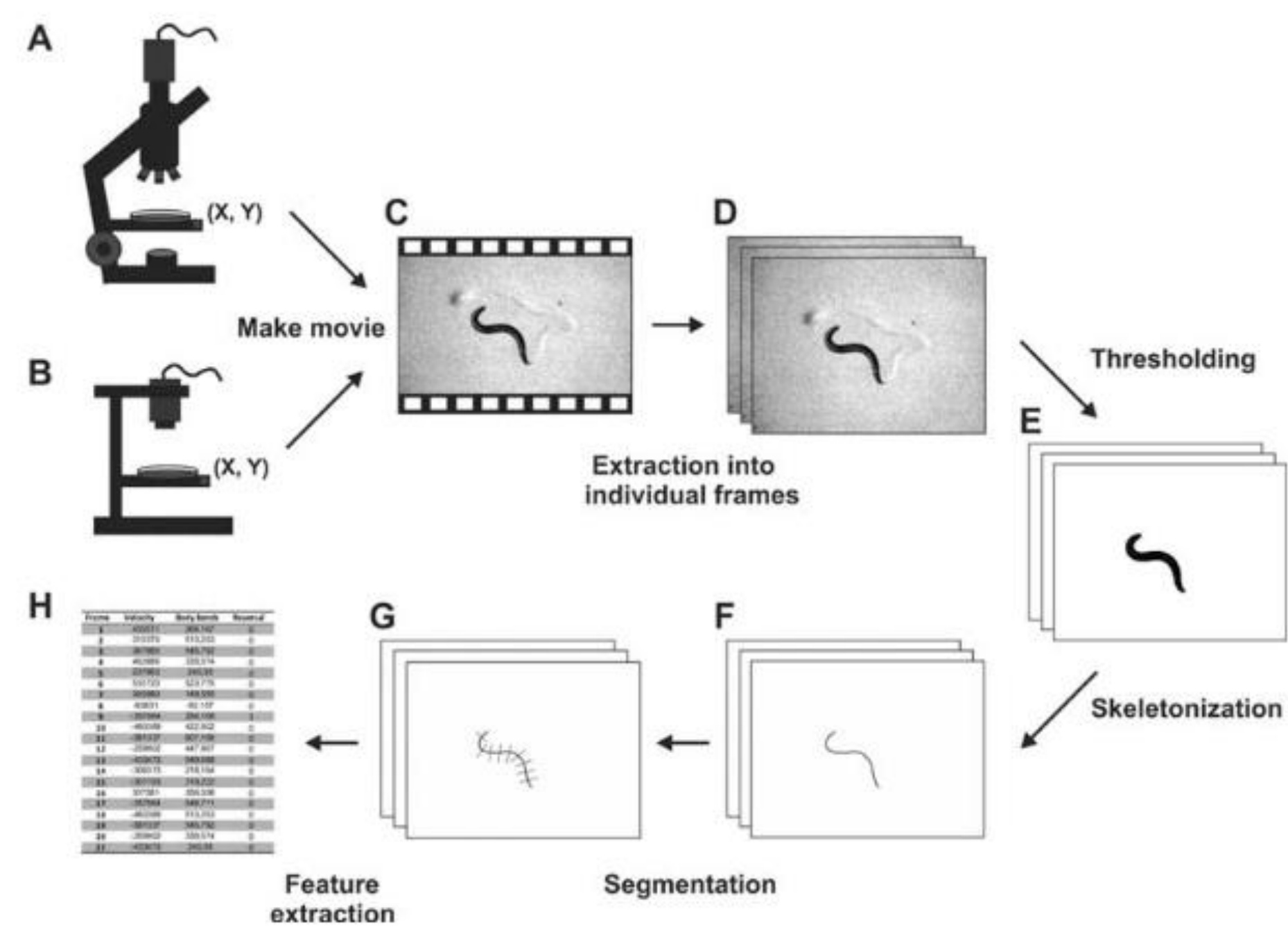
Classification of Behaviors using Posture Features in *C.elegans* Behavioral Database

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Introduction

Nematode C. elegans is a popular model organism for **neurobiological** and **behavioral** aspects.



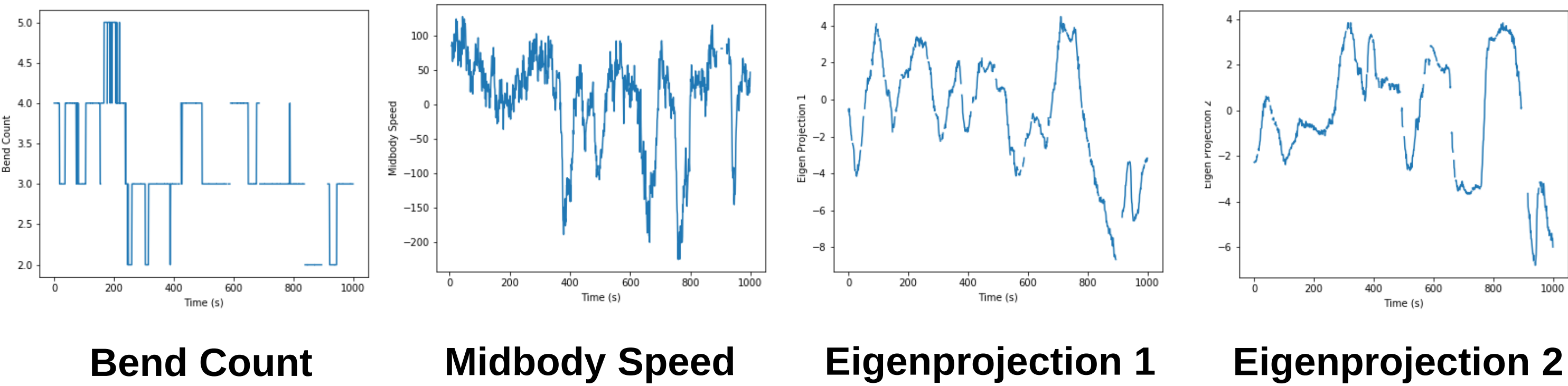
We collect behavioral data of *C. elegans* compiled by automatic imaging techniques during experiments which constitute various **posture features**.

We construct and train ANN model to take **posture features** as inputs and outputs **behavioral types** (e.g. forward, backward, pause).

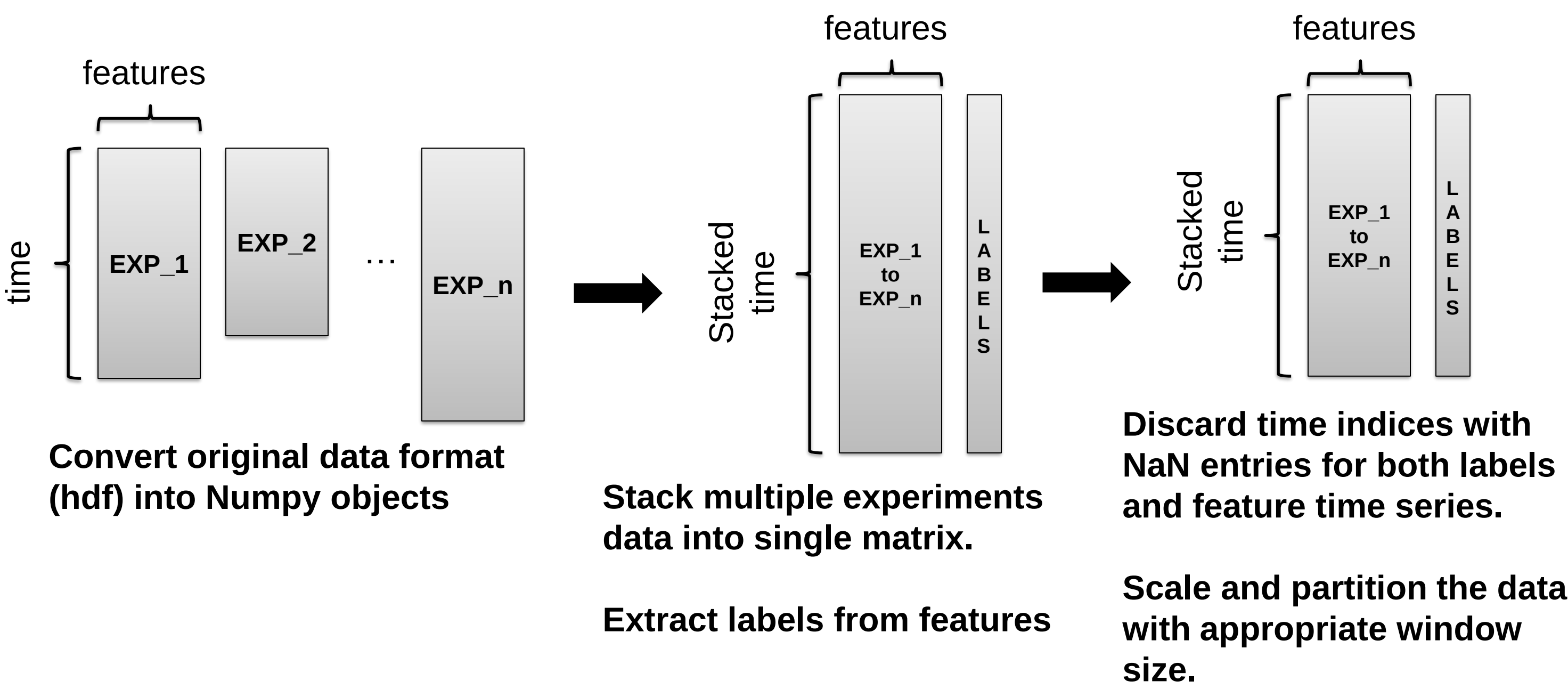
Through this study, we aim to show **how predictive posture features** are for classifying behavior types, with possible application in experiments to assist in **auto-labeling of behaviors**.

C.elegans Behavioral Database

We use data from **OpenWorm behavioral database**: **13,339** experiments with time series of features. Each experiment consists of **~25,000** video frames (~1000s) and **44** posture features. .

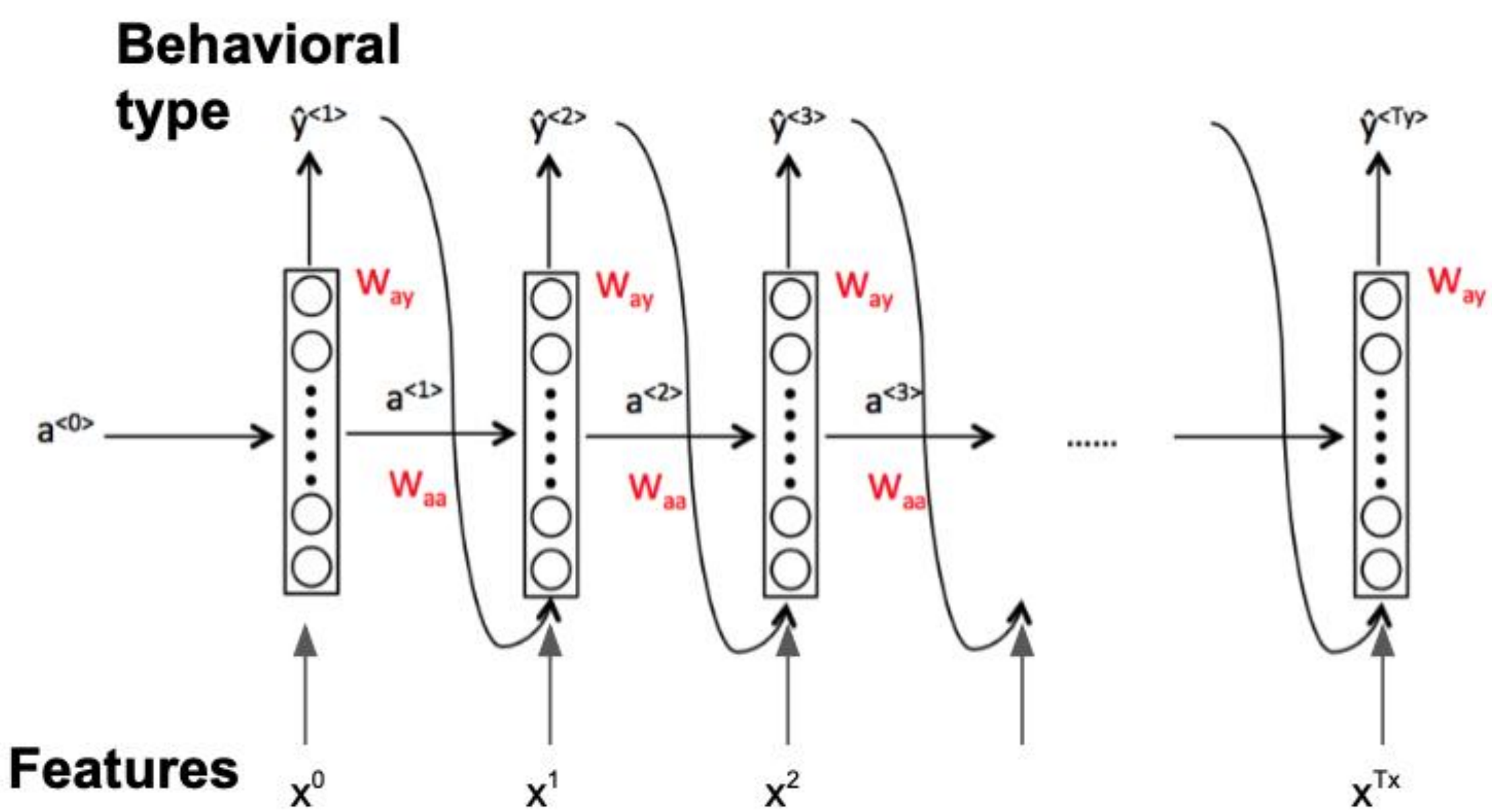


Data Preprocessing and Feature Extraction



ANN Architecture

We use **many to many LSTM** to take time sequence of **posture features** as inputs and outputs **behavioral types** (Forward, Backward, Pause).



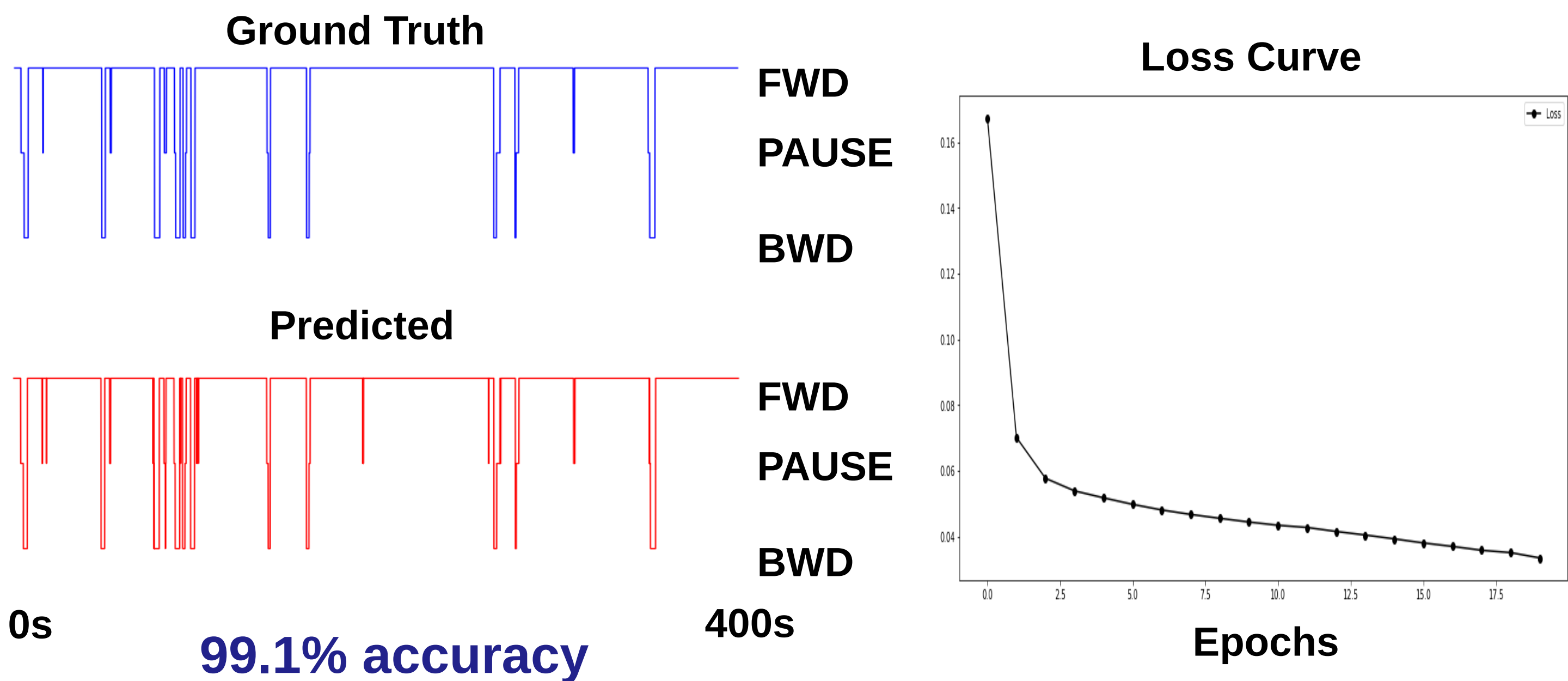
For the results below we use data window size of **50 (2s)**, batch size of **10** and hidden layer size of **512**

Results

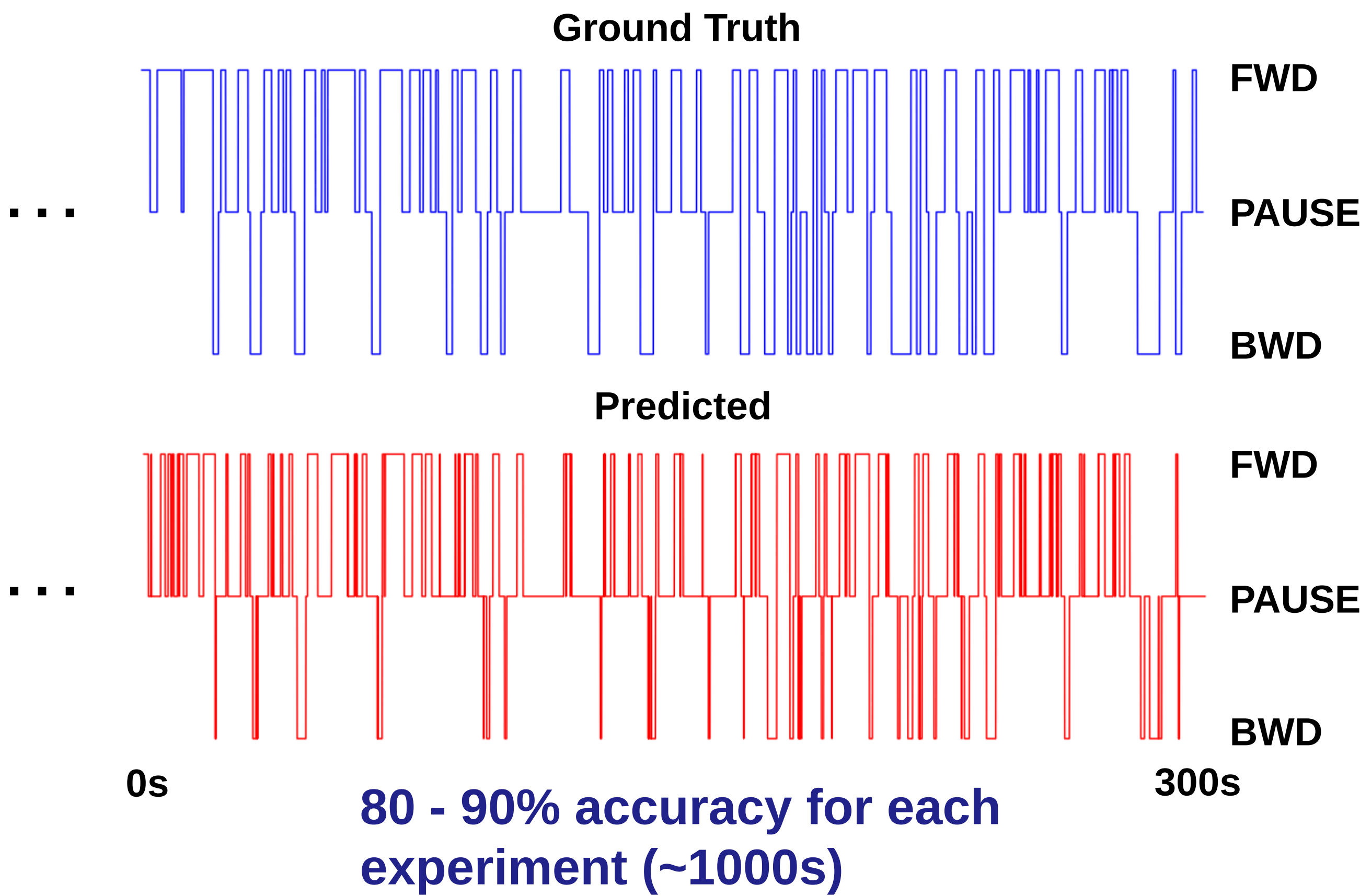
Prediction on Testing Sample

(150 Experiment sets, Lr = 0.0001, 20 Epochs)

(9 Features used: Eigenprojection 1 – 6, head, mid-body, tail absolute speed)



Prediction on Random Experiment Sets



References
[1] Yemini, E., Jucikas, T., Grundy, L. J., Brown, A. E., & Schafer, W. R. (2013). A database of Caenorhabditis elegans behavioral phenotypes. *Nature methods*, 10(9), 877.